HW 7

1) 8.1 12

12) V: 5 $\{v_1, v_2, v_3, v_4, v_5\}$
E: 8 $\{e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8\}$
parallel edges: none
loops: none
isolated vertices: none
G is a simple graph
e_1 is incident on v_4, v_2

13) V: 3 $\{v_1, v_2, v_3\}$
E: 0 $\emptyset$
PE: no
L: no
isolated vertices: 0
G is a simple graph
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2) 8.1 16

16) $K_1 = 0$ $K_2 = 1$ $K_3 = 3$

sets of two vertices: n options for first vertex, n-1 for second.
(u, v) = (v, u) so divide by 2.

$\frac{n(n-1)}{2}$ → # edges in K_n

3) ~~17) bipartite $u = \{v_1, v_2, v_5\}, v = \{v_3, v_4\}$~~
18) yes $u = \{v_3, v_2, v_7\}$ $v = \{v_1, v_6, v_3, v_4, v_8, v_9, v_{10}\}$ bipartite
~~19) yes $u = \{Gre, Buf, Sho, Mud, Dou\}$ $v = \{She, Wac, Lan, Cas, Gil\}$~~
~~20) yes $u = \{v_2, v_3, v_4\}$ $v = \{v_1, v_3, v_6\}$~~
~~21) no~~
~~22) no~~
~~23)~~

4) 8.1 41

edges from each vertex = n
vertices = 2^n (2 choices for each?) /2 to only count edges once

$\frac{n2^n}{2} = \boxed{n2^{n-1}}$

5) 8.2 2) none 3) simple path only 5) cycle only 6) none bc repeated edge & vertices 8) simple path only 9) simple path only

6)

